A Variational Perspective on Flow Matching for VQ-Based Image Synthesis

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Abstract

Generative modeling in discrete latent spaces has become a cornerstone for high-resolution image synthesis. While Flow Matching offers an efficient alternative to diffusion models in continuous domains, its application to discrete categorical distributions such as those produced by a Vector Quantized Variational Autoencoder remains non-trivial. In this work, we adapt the Variational Flow Matching framework, originally proposed for graph generation, to the discrete latent space of images. By formulating the probability paths through a variational lower bound, we enable the model to learn optimal trajectories between noise and data distributions. We parameterize the time-dependent vector field using a Transformer architecture to capture global spatial dependencies. Our results suggest that VFM provides a flexible and efficient generative prior, outperforming standard fixed-path approaches in discrete image synthesis.

1 Introduction

The development of two-stage generative frameworks has advanced high-fidelity image synthesis. By leveraging Vector Quantized Variational Autoencoders (VQ-VAEs) [7], researchers can compress high-dimensional images into a compact grid of discrete latent tokens, reducing the complexity of the subsequent generative task. Traditionally, Autoregressive Transformers or Discrete Diffusion Models have been employed to model these latent priors. However, these methods often face a trade-off between the sequential bottleneck of autoregression and the complex noise schedules of diffusion.

Flow Matching (FM) [5] has recently emerged as a compelling paradigm, providing a simulation-free objective for training continuous-time generative models with straight-line trajectories. Despite its success in continuous spaces, standard FM relies on predefined probability paths that may not capture the underlying structure of categorical data. Variational Flow Matching (VFM) [3] addresses this by introducing a variational framework to learn these probability paths. While VFM was initially tailored for the permutation invariance and topological constraints of graph generation, its application to the structured grids of VQ-VAE latents offers a promising direction for more efficient image generation.

In this report, we propose a discrete VFM prior for image synthesis. We first map images into a discrete latent space and then define a flow-based generative process over the resulting categorical distribution. Our model utilizes a Transformer-based vector field to estimate the probability flow.

2 Related Work

2.1 Flow Matching

Flow matching is a generative modeling framework that transforms a simple distribution into a complex data distribution by learning how points should "flow" through space over time. Instead of jumping directly from noise to data, we define a velocity field $u_t(x)$. This field acts like a map of vectors that tells a point x at time t exactly which direction to move and how fast. By following these vectors—essentially solving an ODE—we can transport a sample x_0 from the starting noise distribution p_0 to a final point x_1 that looks like real data.

Formally, we define

$$\frac{d}{dt}\phi_t(x) = u_t(\phi_t(x)), \quad \phi_0(x) = x \tag{1}$$

where $\phi_t(x) : [0,1] \times \mathbb{R}^D \to \mathbb{R}^D$ is a time-dependent diffeomorphism. By integrating this velocity over time, we transform noise $x_0 \sim p_0$ into a clean sample $\phi_1(x_0) \sim p_{data}$, and we define $p_t(x)$ as the law of $\phi_t(x_0)$.

To train a model, we ideally want to match the true underlying velocity field $u_t(x)$. However, this marginal field is intractable. Fortunately, $u_t(x)$ can be expressed through conditional trajectory $u_t(x|x_1)$ generated by conditional path $p_t(x|x_1)$:

$$u_t(x) = \int u_t(x|x_1)p_t(x_1|x)dx_1 = \int u_t(x|x_1)\frac{p_t(x|x_1)p_{data}(x_1)}{p_t(x)}dx_1 \tag{2}$$

where usually $u_t(x|x_1) = \frac{x_1 - x}{1-t}$. Using this representation one can show that

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{\substack{t\sim[0,1],\, x_1\sim p_{\text{data}} \\ x\sim p_t(x|x_1)}}\left[\left\|v_t^\theta(x) - u_t(x|x_1)\right\|_2^2\right] = \mathbb{E}_{t,x\sim p_t}\left[\left\|v_t^\theta(x) - u_t(x)\right\|_2^2\right] + C \tag{3}$$

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2.2 Variational Formulation

Rather than treating the problem as a direct regression of the conditional velocity, we can interpret the objective through the lens of posterior alignment. By leveraging the relationship defined in (2), the optimization can be reformulated as minimizing the cumulative KL divergence between the true (but intractable) posterior $p_t(x_1|x)$ and a parameterized variational approximation $q_t^\theta(x_1|x)$:

$$\mathcal{L}_{\text{VFM}}(\theta) = \int_0^1 \mathbb{E}_{x\sim p_t(x)}\left[D_{\text{KL}}(p_t(x_1|x) \parallel q_t^\theta(x_1|x))\right] dt \tag{4}$$

By expanding the definition of the KL divergence and removing the entropy term of the data (which is constant with respect to θ), we arrive at a tractable surrogate objective. This yields a formulation centered on the expected log-likelihood of the target data:

$$\mathcal{L}_{\text{VFM}}(\theta) = -\mathbb{E}_{t\sim[0,1],x_1\sim p_{\text{data}},x\sim p_t(x|x_1)}\left[\log q_t^\theta(x_1|x)\right] + C \tag{5}$$

where C is a constant. This reveals that Flow Matching can be viewed as maximizing the probability of the endpoint x_1 given an intermediate noisy state x at any time t.

2.3 Mean-Field Factorization

By leveraging the linearity of the conditional vector field $u_t(x|x_1)$ in (2), we can express the marginal velocity field as a function of the posterior mean:

$$u_t(x) = \mathbb{E}_{x_1\sim p_t(x_1|x)}\left[\frac{x_1 - x}{1-t}\right] = \frac{\mathbb{E}_{x_1\sim p_t(x_1|x)}[x_1] - x}{1-t}. \tag{6}$$

Since the posterior mean $\mathbb{E}_{x_1\sim p_t(x_1|x)}[x_1]$ is a D-dimensional vector composed of independent scalar expectations for each dimension $d \in \{1, \ldots, D\}$, we can simplify the variational distribution $q_t^\theta(x_1|x)$. Specifically, by adopting a Mean-Field approximation, we assume the dimensions of x_1 are conditionally independent:

$$q_t^\theta(x_1|x) = \prod_{d=1}^D q_t^\theta(x_1^d|x). \tag{7}$$

Under this assumption, the Variational Flow Matching (VFM) objective in (5) decomposes into a sum of independent log-likelihood terms for each dimension:

$$\mathcal{L}_{\text{VFM}}(\theta) = -\mathbb{E}_{t\sim[0,1],x_1\sim p_{\text{data}},x\sim p_t(x|x_1)}\left[\sum_{d=1}^D \log q_t^\theta(x_1^d|x)\right] + C. \tag{8}$$

This factorization significantly reduces the complexity of the model, allowing the neural network to focus on modeling the marginal distributions of each feature while still capturing the global flow dynamics through the shared dependence on x and t.

2.4 Distillation

Standard Flow Matching efficiently models continuous-time trajectories, but simulating these trajectories during inference requires numerous numerical ODE solver steps, which is computationally expensive. To accelerate sampling, recent works have introduced *shortcut models* [4] and *ShortCutting Flow Matching* (SCFM) [2]. The core intuition behind these methods is trajectory self-consistency: a model should produce the same output whether it takes one large step of size $d_1 + d_2$ or two smaller consecutive steps of sizes d_1 and d_2.

In the continuous domain, this is enforced by penalizing the discrepancy between the velocity predicted for a large leap and the weighted combination of velocities from smaller steps. SCFM, for instance, distills a pre-trained flow matching model into a few-step sampler by minimizing a Mean Squared Error (MSE) over these velocity vectors.

3 Methodology

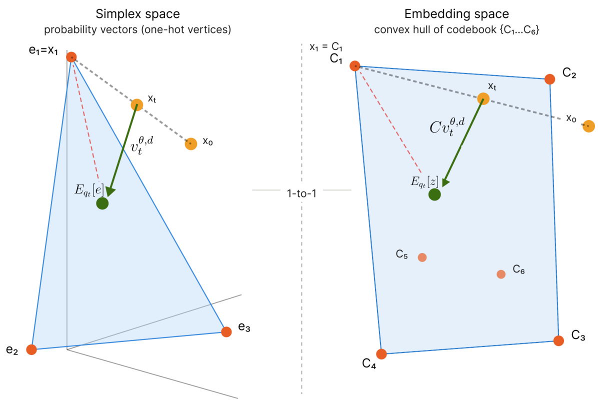

Figure 1: The left panel shows the original variational flow matching framework operating in the probability simplex Δ^{k-1}, where x_0 lies outside the simplex and the linear interpolation path between x_0 and x_1. The right panel shows the proposed reformulation in the codebook embedding space, where the predicted $\sum_i q_t^\theta(x_1^d = k|x_t)z_k$ lives inside the convex hull of the codebook embeddings, and the velocity field v_t^θ points from the noisy sample x_t toward this mean, which progressively moves toward the target $x_1 = C_1$.

3.1 Curse of Dimensionality

Let C be a codebook matrix of size $\mathbb{R}^{E\times K}$, where E is the size of embeddings and K is the codebook size, e_k be the vector of zeros of size K with 1 at k_{th} place, $z_k \in \mathbb{R}^E$ be the k_{th} column of C, $x_1 = (x_1^1, \dots, x_1^D)^T$ be the vector of tokens for input to decoder, where D is the sequence length, $x_1^d \in \{1, \dots, K\}$. Note that x_1^d can be represented as a One-Hot vector e_k for some k. Then estimated marginal velocity $v_t^\theta(x)$ though equation (8) has the form:

$$v_t^{\theta,d}(x) = \sum_{k=1}^{K} q_t^\theta(x_1^d = k|x)\frac{e_k - x^d}{1-t}, \text{ for } d \in 1, \dots, D \tag{9}$$

A significant practical challenge arises from the stochastic interpolation $x_t^d = te_k + (1-t)x_0^d$. If we assume a standard Gaussian noise initialization $x_0^d \sim \mathcal{N}(0, I_K)$, the expected squared norm (energy) of the state at time t is:

$$\mathbb{E}[\|x_t^d\|^2] = t^2\|e_k\|^2 + (1-t)^2\mathbb{E}[\|x_0^d\|^2] = \mathbb{E}[\|x_t^d\|^2] = t^2 + (1-t)^2K \tag{10}$$

for different values of K as t goes from 0 to 1. For large codebook sizes K, this formulation leads to signal-to-noise ratio collapse. At the start of the trajectory ($t \approx 0$), the energy is dominated by the noise term K. The informative signal (the one-hot target) is suppressed by a factor of $1/K$. This high-variance environment forces the model to recover a needle (the vertex e_k) from an K-dimensional haystack of noise, making early-time gradients extremely unstable. Another issue is computational inefficiency. Storing and computing vector fields in $\mathbb{R}^{D\times K}$ is memory-prohibitive. For a sequence of length $D = 512$ and $K = 8192$, a single sample x_t requires ≈ 4.2 million parameters. This bypasses the compact representation offered by the latent embedding space $\mathbb{R}^E$, where typically $E \ll K$.

As we can observe, the standard CatFlow approach ignores the underlying geometry of the codebook C. By treating classes as equidistant vertices in $\mathbb{R}^K$, we incur a "dimensionality tax" that scales linearly with the vocabulary size, rendering the method impractical for high-resolution latent spaces.

3.2 Diffusion in Latent Space

To mitigate the curse of dimensionality, we project the categorical dynamics into the latent embedding space $\mathbb{R}^E$. By linearity of the expectation and the codebook mapping $z_k = Ce_k$, multiplying the velocity equation by the codebook matrix C yields:

$$\mathbf{f}_t^{\theta,d} = Cv_t^{\theta,d}(x_t) = \sum_{k=1}^{K} q_t^\theta(x_1^d = k|x_t)\frac{z_k - Cx_t^d}{1-t} = \frac{\mathbb{E}q_t^\theta[z] - \mathbf{z}_t^d}{1-t} \tag{11}$$

where $\mathbf{z}_t^d = Cx_t^d \in \mathbb{R}^E$ and the latent velocity $\mathbf{f}_t^{\theta,d} = Cv_t^{\theta,d}$. The last step is the transition from $q_t^\theta(x_1^d = k|x_t)$ to $q_t^\theta(x_1^d = k|z_t)$ which requires

Assumption: The latent state $z_t^d = Cx_t^d$ acts as a sufficient statistic for the target distribution.

Since x_t is a noisy interpolation in $\mathbb{R}^K$ and z_t is its linear projection via the codebook matrix C, we are effectively defining a mapping $\Phi : \mathbb{R}^K \to \mathbb{R}^E$. We argue that:

$$q_t(x_1^d|x_t) \approx q_t(x_1^d|\Phi(x_t)) = q_t(x_1^d|z_t) \tag{12}$$

This approximation is valid if the codebook C is well-conditioned. If the columns of C (the embeddings z_k) are distinct and span the relevant features of the data, the projection into $\mathbb{R}^E$ preserves the "directional" information needed to identify the target token x_1^d, while discarding the K-dimensional noise that does not align with the codebook subspace.

In the original space $\mathbb{R}^K$, the model must learn a vector field over a massive, mostly empty volume. By switching to z_t, we constrain the model to learn the distribution relative to the geometry of the embeddings.

The model now operates in $\mathbb{R}^E$ instead of $\mathbb{R}^K$ (see Fig. 1 for comparison). Since $E \ll K$ (e.g., 512 vs 8192), the memory footprint and computational cost of the ODE integration are reduced by orders of magnitude. As the expected norm in latent space $\mathbb{E}[\|\mathbf{z}_t\|^2]$ now depends on the scale of the embeddings in C rather than the vocabulary size K, the "energy explosion" at $t \approx 0$ is prevented which leads to more stable training gradients. Finally, by working with z_k, the model implicitly leverages the semantic structure of the codebook. Errors between tokens with similar embeddings result in smaller loss values, providing a smoother optimization landscape compared to the discrete simplex. Plugging (12) into (6) we immediately get training objective in the latent space. For generation, we integrate $\mathbf{f}_t^{\theta,d}$ and search for the closest z_k in a codebook.

3.3 Training via Shortcut Variational Flow Matching (SVFM)

This section adapts continuous velocity matching into a discrete probability matching objective for categorical frameworks. The approach relies on two core mechanisms:

The Probability Mixture: When translating continuous two-step velocity matching into the latent formulation, the noisy state mathematically cancels out. This simplifies the student's target into a direct convex mixture of the teacher's categorical probability distributions. The student is then trained using a standard KL divergence.

Dual-EMA Bootstrapping: To stabilize the training dynamics and avoid moving-target problems, two Exponential Moving Average (EMA) copies of the student model are maintained: a Fast EMA (acting as an aggressive local teacher) and a Slow EMA (acting as a conservative global anchor). Training batches use a combination of pure distillation from a frozen base model and self-consistency guidance from these EMA models.

Please refer to the appendix E for detailed mathematical derivations.

4 Experiments

In this section, we detail our experimental setup and evaluate the performance of our Variational Flow Matching (VFM) prior on the CIFAR-10 dataset. Our implementation is built upon the LlamaGen framework [6], adapted for the VFM objective (see Appendix B for training setup).

4.1 Generative Results

The performance of our VFM model is evaluated using the Fréchet Inception Distance (FID) on 10,000 generated samples. Our model achieves a final **FID-10K of 21.2**. As shown in Figure 8, the model successfully synthesizes recognizable images with consistent global structures, validating the effectiveness of learning probability paths in the discrete latent space.

The sampling process, facilitated by the learned flow trajectories, demonstrates that VFM can capture the categorical distribution of image tokens effectively. While autoregressive models often suffer from error accumulation, the

Table 1: Quantitative results on CIFAR-10.

Model	Parameters	rFID-10K ↓	FID-10K ↓
VQ-VAE (Ours)	16M	5.47	-
VFM-Transformer (Ours)	110M	-	**21.2**

flow-based approach provides a more holistic transition from noise to data. However, the SVFM proved to be ineffective and yielded poor generative results, despite the tweaks and adjustments attempted. The root of this issue, named off-manifold stepping, is detailed in appendix F.

4.2 Closed-form

In this section, we try to reproduce the experiments from [1].
The first experiment is to compute the average error between the learnt velocity field $v^\theta(x_t, t)$ and the **closed-form** velocity field $v^*(x_t, t) = \frac{q^*(x_t,t)-x_t}{1-t}$ where $q*$ is the exact solution to the CatFlow optimization problem.

Proposition 1 (Analytic form of q^*). *For a dataset $\mathcal{D}_n = \{x_i : 1 \le i \le n\} \subset \{e_1, \dots, e_K\}^D$ and a prior distribution p_0, the optimal solution $q^*(x_t, t)$ is given by :*

$$q^*(x_t, t) = \sum_{i=1}^{n} \lambda_i(x_t, t)\, x_i \quad \textit{where} \quad \lambda_i(x, s) = \frac{p_0\left(\frac{x - s x_i}{1-s}\right)}{\sum_{j=1}^{n} p_0\left(\frac{x - s x_j}{1-s}\right)} \forall i \in \{1, \dots, n\}$$

Proof (see Section A.1). ... □

Note that the closed-form velocity $v^*(x_t, t)$ is therefore the same as with the CFM loss.

To evaluate this average error, we trained a bi-directional transformer on a smaller dataset **Uniprot**. After training, we defined a regular grid of timesteps t, we sampled 256 times $x0 \sim p_0$ and $x_1 \sim \hat{p}_{data}$, and we computed the average norm : $e(t) = \mathbb{E}_{x_0, x_1}\left[\|v^\theta(x_t, t) - v^*(x_t, t)\|_2\right]$.
Results for CatFlow objective are given in figure 2, and results for CFM objective are given in figure 3. We use either uniform prior on simplex – defined as $x_0 \sim p_0^{uni}(x) = \prod_{d=1}^{D} C_K \mathbf{1}[x^{(d)} \in \Delta^{K-1}]$ – or Gaussian prior (actually a scaled-gaussian prior) – defined as $x_0 \sim p_0^g(x) = \prod_{d=1}^{D} \mathcal{N}(x^{(d)} | 0, \sigma^2 I_K)$ with the scale $\sigma = \sqrt{K}$.

We see that for both objectives, the error jumps dramatically for small timesteps ($t \approx 0.1$ for uniform prior, $t \approx 0.2$ for gaussian prior), and then stabilizes near 0 after a certain time τ (before exploding near $t = 1$). We can see that the time for "error peak" correlates counter-intuitively with the time when closed-form velocity and conditional velocity are completely aligned (see figure 4), i.e. the time when v^* should be very simple to learn. This failure to learn optimal velocity can be explained by a very high Lipschitz constant of v^* with respect to x near $t \approx t_{failure}$. This phenomenon is consistent with the analysis in [1].

An other interesting observation is that *memorization* seems to be dependent on prior and objective (figure 5). Only models trained with Gaussian prior are able to generate samples that are present in the training set. CatFlow objective seems to accentuate this tendency compared to CFM.
We know that, depending on the prior, the failure time $t_{failure}$ happens sooner or later. The later failure happens, the less it has impact on the overall trajectory. For Gaussian prior, the failure happens sufficiently late, so that some trajectories can nevertheless converge to a data point.

5 Conclusion

In this work, we have presented a novel adaptation of the Variational Flow Matching framework for generative modeling in discrete latent spaces. By bridging the gap between categorical probability paths and vector-quantized image representations, we have addressed the fundamental "curse of dimensionality" that typically plagues flow-based models in large-vocabulary settings. Our theoretical analysis demonstrated that operating directly on the probability simplex Δ^{K-1} leads to a catastrophic Signal-to-Noise Ratio collapse and an "energy explosion" as the vocabulary size K increases. To resolve this, we introduced a latent projection method that treats the codebook embedding space as a sufficient statistic for the categorical distribution. This transition not only stabilizes the training dynamics by providing a bounded energy profile but also allows the model to leverage the underlying semantic geometry of the codebook. As the result, we achieved significant gains in computational efficiency and memory optimization without sacrificing generative fidelity. Experimentally, our Transformer-based VFM prior demonstrated the ability to synthesize high-quality images on CIFAR-10, achieving a FID-10K of 21.2. Furthermore, our investigation into the closed-form velocity fields revealed that while learning trajectories at very small timesteps remains a challenge due to high initial variance, the VFM objective successfully aligns with the optimal probability paths early in the generative process. Finally, exploring SVFM revealed that continuous ODE distillation fails in categorical spaces due to off-manifold intermediate states. Given VFM's native few-step efficiency, future acceleration efforts should prioritize inherently discrete paradigms. We believe that the integration of variational flows with discrete representation learning constitutes a robust and flexible paradigm for the next generation of generative models.

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A Detailed Proofs

A.1 Closed form for CatFlow loss

One-dimensional case We denote as Δ^{K-1} the $(K-1)$-dimensional simplex, i.e. the set of K-dimensional vectors with non-negative entries that sum to 1.
We denote as e_k the k-th standard basis vector in $\mathbb{R}^K$.
Let $\mathcal{D}_n = \{x_i : 1 \leq i \leq n\} \subset \{e_1, \ldots, e_K\}$ denote the dataset.
Let $\hat{p}_{data} = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$ denote the empirical distribution of the dataset.
Let $p_0 \in \mathcal{P}(\mathbb{R}^K)$ denote the distribution of the initial state x_0.

Let $X_1 \sim \hat{p}_{data}$, $X_0 \sim p_0$, $t \sim \mathcal{U}([0,1])$. Let $X_t = (1-t)X_0 + tX_1$.
We denote as p_t the distribution of the state x_t at time t.
Let $q : \mathbb{R}^K \times [0,1] \to \Delta^{K-1}$ denote a velocity field.

The CatFlow loss is a cross-entropy loss, defined as:

$$\mathcal{L}(q) = -\mathbb{E}\left[\sum_{k=1}^K X_1^{(k)} \log q^{(k)}(X_t, t)\right] \tag{13}$$

The optimization problem is to find the velocity field q that minimizes the loss:

$$\min_q \mathcal{L}(q) \tag{14}$$

By conditioning on X_t and t, we can rewrite the loss as:

$$\mathcal{L}(q) = -\mathbb{E}\left[\sum_{k=1}^K \mathbb{E}[X_1^{(k)}|X_t, t] \log q^{(k)}(X_t, t)\right] \tag{15}$$

We know that q_* minimizes the loss if and only if for all $x \in \mathbb{R}^K$ and $s \in [0,1]$, we have:

$$q_*(x, s) = \mathbb{E}[X_1 | X_t = x, t = s] \tag{16}$$

We also know that :

$$p_t(x|X_1 = x_i) = \frac{1}{(1-t)^K} p_0\left(\frac{x - tx_i}{1-t}\right) \tag{17}$$

Thus, by Bayes' rule, we have:

$$\lambda_i(x, s) = \mathbb{P}[X_1 = x_i | X_t = x, t = s] = \frac{p_0\left(\frac{x - sx_i}{1-s}\right) \hat{p}_{data}(x_i)}{\sum_{j=1}^n p_0\left(\frac{x - sx_j}{1-s}\right) \hat{p}_{data}(x_j)} = \frac{p_0\left(\frac{x - sx_i}{1-s}\right)}{\sum_{j=1}^n p_0\left(\frac{x - sx_j}{1-s}\right)} \tag{18}$$

Hence, we have:

$$\boxed{q_*(x, s) = \sum_{i=1}^n \lambda_i(x, s)\, x_i} \tag{19}$$

Multi-dimensional case Let D be the sequence length, and K be the number of classes.
Let $\mathcal{D}_n = \{x_i : 1 \leq i \leq n\} \subset \{e_1, \ldots, e_K\}^D$ denote the dataset.
Let $\hat{p}_{data} = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$ denote the empirical distribution of the dataset.
Let $p_0 \in \mathcal{P}((\mathbb{R}^K)^D)$ denote the distribution of the initial state x_0.
Let $X_1 \sim \hat{p}_{data}$, $X_0 \sim p_0$, $t \sim \mathcal{U}([0,1])$. Let $X_t = (1-t)X_0 + tX_1$.
We denote as p_t the distribution of the state x_t at time t.
Let $q : ((\mathbb{R}^K)^D) \times [0,1] \to (\Delta^{K-1})^D$ denote a velocity field.

The CatFlow loss is a cross-entropy loss **independently applied to each position in the sequence**, defined as:

$$\mathcal{L}(q) = -\mathbb{E}\left[\sum_{d=1}^D \sum_{k=1}^K X_1^{(d,k)} \log q^{(d,k)}(X_t, t)\right] \tag{20}$$

The optimization problem is to find the velocity field q that minimizes the loss:

$$\min_q \mathcal{L}(q) \tag{21}$$

As in the one-dimensional case, by conditioning on X_t and t, we can rewrite the loss as:

$$\mathcal{L}(q) = -\mathbb{E}\left[\sum_{d=1}^{D}\sum_{k=1}^{K}\mathbb{E}\left[X_1^{(d,k)} \mid X_t, t\right]\log q^{(d,k)}(X_t, t)\right] \tag{22}$$

Since the loss is a sum of cross-entropies over positions, the minimizer is obtained pointwise by matching each categorical distribution to the corresponding conditional expectation. Therefore, for all $x \in ((\mathbb{R}^K)^D)$ and $s \in [0, 1]$, we have:

$$q_*(x, s) = \mathbb{E}[X_1 \mid X_t = x, t = s] \tag{23}$$

or equivalently, for each position $d \in \{1, \ldots, D\}$,

$$q_*^{(d)}(x, s) = \mathbb{E}[X_1^{(d)} \mid X_t = x, t = s]. \tag{24}$$

Now let $x_i \in \{e_1, \ldots, e_K\}^D$ denote the i-th sequence in the dataset. Since

$$X_t = (1 - t)X_0 + tX_1, \tag{25}$$

conditioning on the event $X_1 = x_i$ gives

$$X_t = (1 - t)X_0 + tx_i. \tag{26}$$

Hence, for $t \in [0, 1)$,

$$p_t(x \mid X_1 = x_i) = \frac{1}{(1-t)^{DK}}\, p_0\!\left(\frac{x - tx_i}{1 - t}\right), \tag{27}$$

where the factor $(1 - t)^{-DK}$ comes from the affine change of variables on the ambient product space, and is independent of i.

Therefore, by Bayes' rule,

$$\lambda_i(x, s) := \mathbb{P}[X_1 = x_i \mid X_t = x, t = s] = \frac{p_0\!\left(\frac{x - sx_i}{1-s}\right)\hat{p}_{data}(x_i)}{\sum_{j=1}^{n} p_0\!\left(\frac{x - sx_j}{1-s}\right)\hat{p}_{data}(x_j)}. \tag{28}$$

Since $\hat{p}_{data}(x_i) = \frac{1}{n}$ for all i, this simplifies to

$$\lambda_i(x, s) = \frac{p_0\!\left(\frac{x - sx_i}{1-s}\right)}{\sum_{j=1}^{n} p_0\!\left(\frac{x - sx_j}{1-s}\right)}. \tag{29}$$

We finally obtain the closed-form solution:

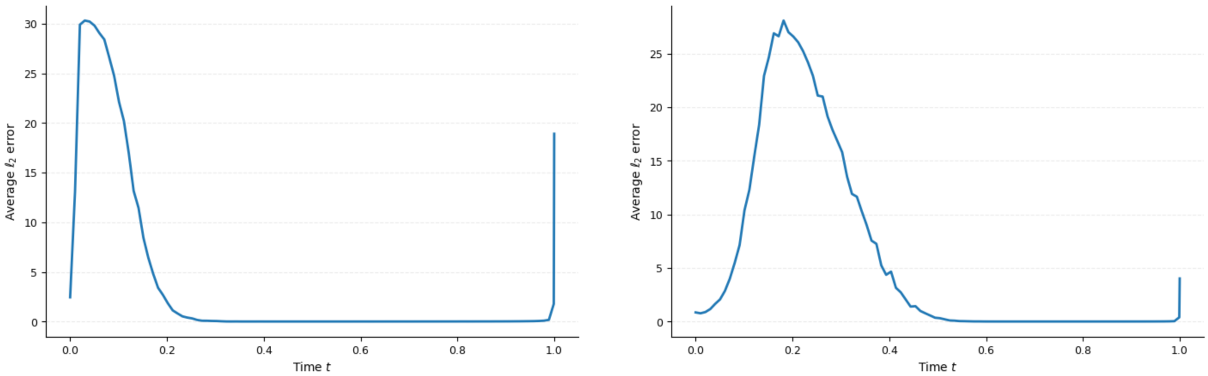
$$\boxed{q_*(x, s) = \sum_{i=1}^{n}\lambda_i(x, s)\, x_i} \tag{30}$$

Equivalently, at each position d,

$$q_*^{(d)}(x, s) = \sum_{i=1}^{n}\lambda_i(x, s)\, x_i^{(d)}, \tag{31}$$

Thus, exactly as in the one-dimensional case, the optimal predictor is a barycenter of the dataset points, with posterior weights given by the probability that each training sequence x_i generated the observation x at time s. One could think that the solution would end up being a product of independent categorical distributions at each position, but this is only the case if the initial distribution p_0 and the data distribution $\hat{p}_{data}$ both factorize as a product of independent distributions at each position.

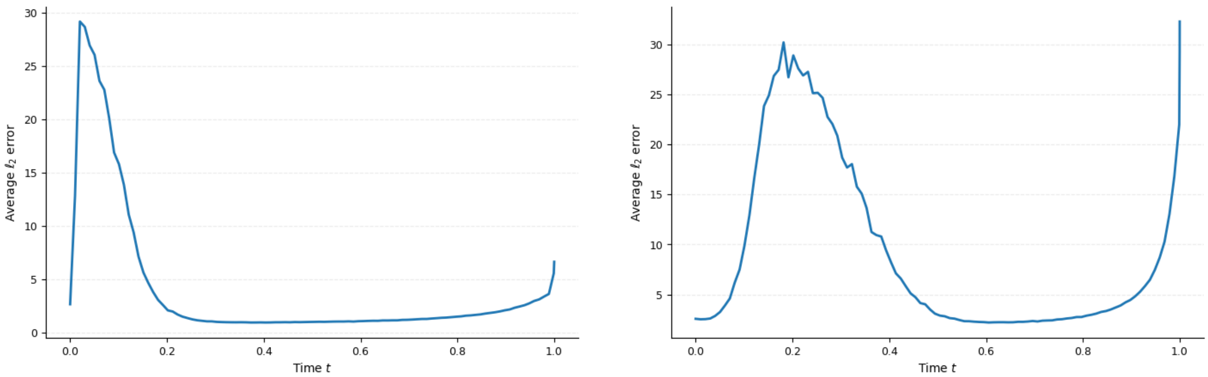
Experimental details We load Uniprot dataset from the official database `https://ftp.uniprot.org/pub/databases/uniprot/current_release/knowledgebase/complete/`, then keep only sequences of minimum length 32, and truncate sequences to the 32 first amino acids. From this filtered dataset, we keep the first 9,000 examples for the training set. There are 20 types of amino acids in the dataset. Our model therefore sees sequences of shape (32, 20).
We train a bi-directional transformer with 8 heads, 8 layers and hidden dimension 512. We train for 300 epochs with batch size 128 (21000 steps).



(a) Uniform prior

(b) Gaussian prior

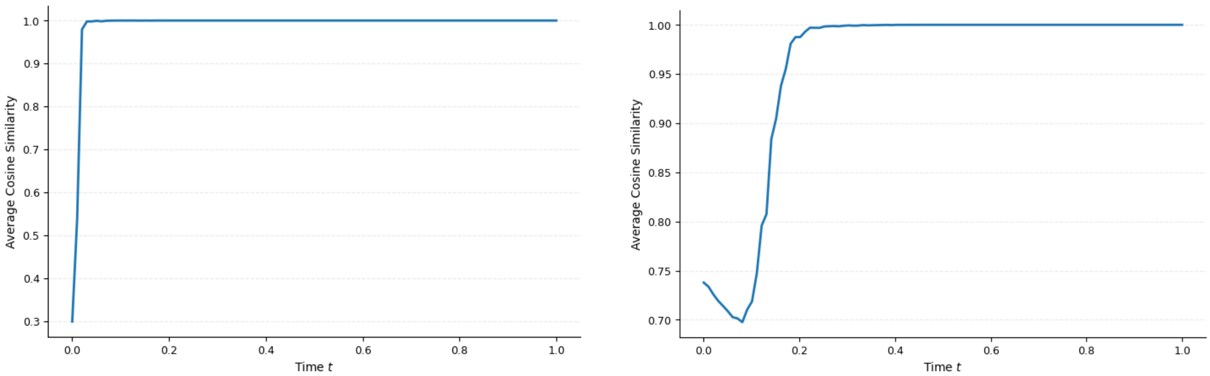
Figure 2: Average L^2 velocity error on Uniprot dataset using CatFlow objective.



(a) Uniform prior

(b) Gaussian prior

Figure 3: Average L^2 velocity error on Uniprot dataset using CFM objective.



(a) Uniform prior

(b) Gaussian prior

Figure 4: Average cosine similarity between closed-form velocity v^* and conditional velocity $x_1 - x_0$ on Uniprot dataset.

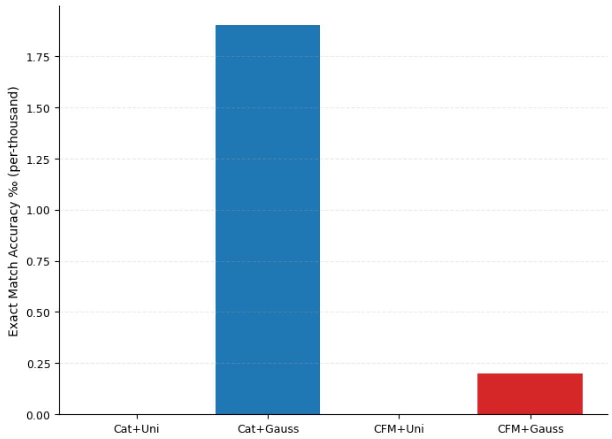


Figure 5: Number of generated samples that fall into the training dataset per thousand

Figures We report here the figures obtained after Uniprot training.
In figure 2 and 3, the failure near $t = 1$ is expected, as v^* becomes non-Lipschitz at $t = 1$.

B Training Setup

We adopt a two-stage generative approach. First, we train a VQ-VAE from scratch on CIFAR-10. To maintain a high-fidelity latent representation, we use a downsampling factor of 2 and a patch size of 2, resulting in a latent grid of 16×16 tokens ($N = 256$). The VQ-VAE achieves a reconstruction FID (rFID-10K) of **5.47**, ensuring that the latent space preserves sufficient structural information for the prior model.

For the generative prior, we employ a Transformer architecture with 110M parameters. The model is trained for 200 epochs with AdamW, learning rate at 1e-4 and weight decay at 0.05. Class conditioning is implemented by prepending a learnable class embedding token to the sequence of latent tokens. This allows the self-attention mechanism to globalize the class information across the spatial grid during the flow process.

To accelerate inference, we apply SVFM to the pre-trained prior. 60% of each batch is guided by the frozen teacher (distillation), while 40% enforces self-consistency using the EMA models. We target an 8-step generation process. To maintain numerical stability and correctly penalize the distributional distance in the categorical latent space, the student is optimized using the KL divergence against the target probability mixtures. We use AdamW with a learning rate of 5×10^{-5} and a cosine annealing scheduler. To focus capacity on critical high-noise regions, a timestep shift $s \in [1.0, 1.2]$ is applied. Final evaluation is performed exclusively on the slow EMA model, which provides the most stable representation of the learned shortcut paths.

C Additional Experimental Results

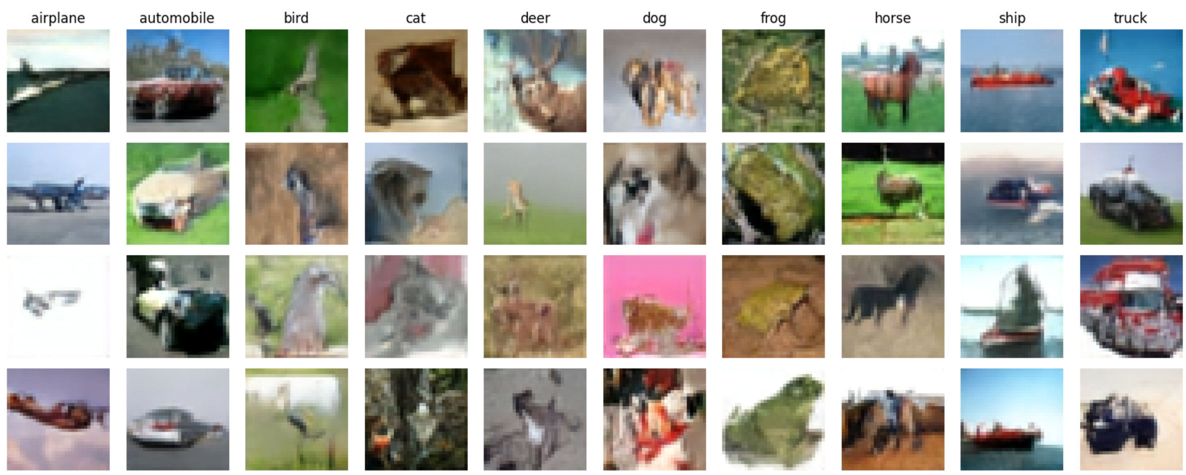


Figure 6: Synthesized 32×32 images generated by our model trained on CIFAR-10 for 150 epochs.

Figure 7: Synthesized 32 $\times$ 32 images generated by our model trained on CIFAR-10 for 200 epochs.

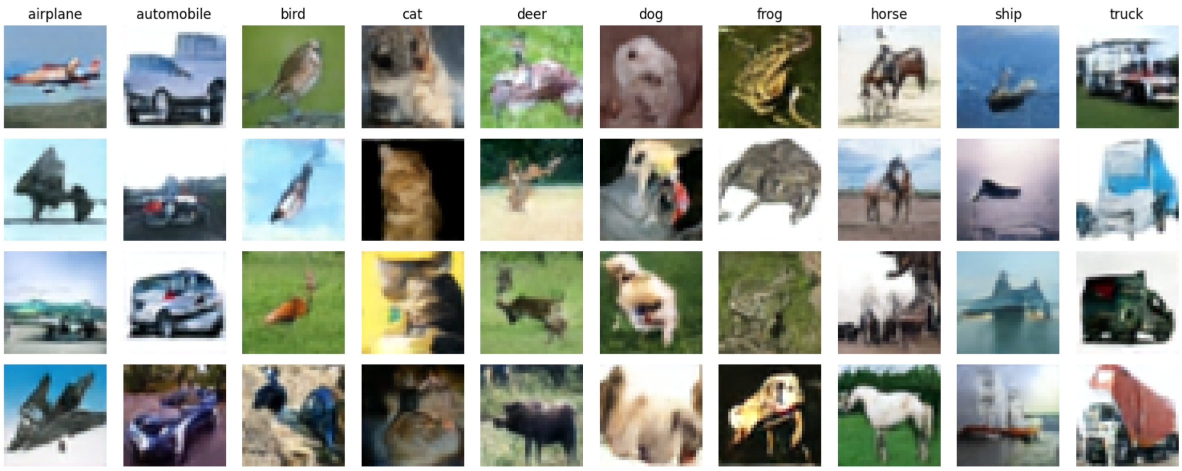


Figure 8: Synthesized 32 $\times$ 32 images generated by our model trained on CIFAR-10 for 200 epochs.

D Training loss

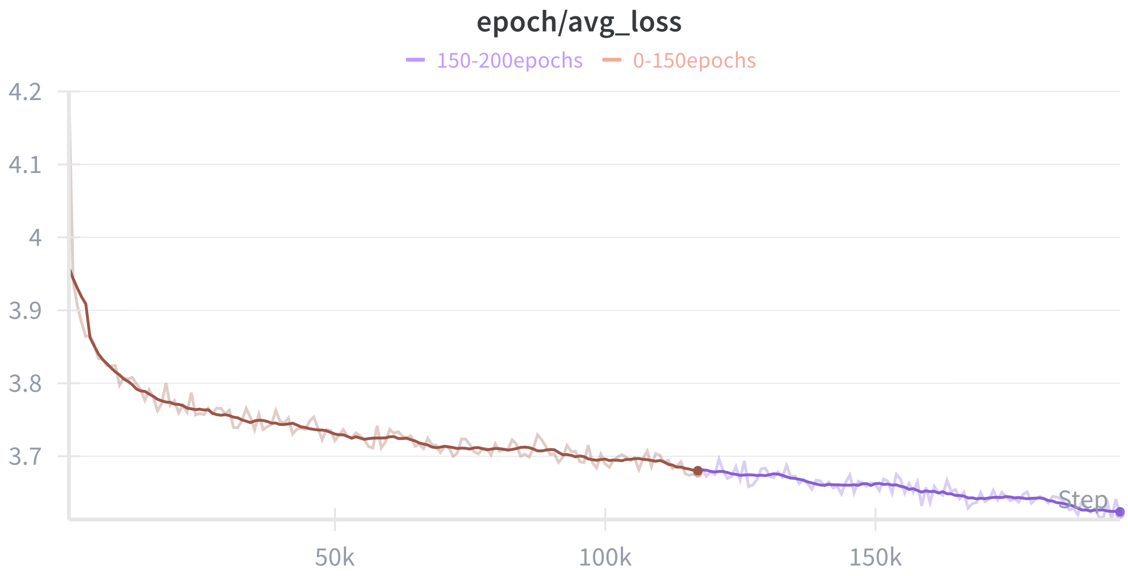


Figure 9: Final loss

E Training via Shortcut Variational Flow Matching (SVFM)

Applying the continuous shortcut logic developed in SCFM directly to discrete, codebook-based latent spaces presents a fundamental "Categorical Gap". In a VQ-VAE latent space, the true target is a categorical distribution over discrete tokens. Applying an MSE loss to regularize intermediate continuous embeddings forces the model to predict spatial averages of codebook vectors, which often correspond to out-of-distribution noise rather than meaningful semantic mixtures. Consequently, the continuous shortcut methodology must be rigorously adapted to operate on probability distributions rather than raw spatial vectors.

To adapt the self-consistency principles of SCFM to our categorical framework, we must translate the continuous velocity matching objective into a discrete probability matching objective. We refer to this as Shortcut Variational Flow Matching.

E.1 The Probability Mixture

Consider three ascending timesteps $t_1 < t_2 < t_3$ transitioning from noise ($t = 0$) to data ($t = 1$), with interval sizes $d_1 = t_2 - t_1$ and $d_2 = t_3 - t_2$. In continuous shortcut distillation, the student's velocity v_S at t_1 over the total interval $d_1 + d_2$ is constrained to match the interpolated velocities of a teacher taking two smaller steps:

$$(d_1 + d_2)v_S(z_{t_1}, t_1) = d_1 v_1(z_{t_1}, t_1) + d_2 v_2(z_{t_2}, t_2) \tag{32}$$

where z_{t_2} is obtained via an Euler step from the first velocity: $z_{t_2} = z_{t_1} + d_1 v_1(z_{t_1}, t_1)$.

In our latent VFM formulation, the velocity is induced by the expected data embedding $\mu_t = \sum_k q_t^\theta(k|z_t) z_k$, yielding $v(z_t, t) = \frac{\mu_t - z_t}{1-t}$. Substituting this definition into Equation 32, we get:

$$(d_1 + d_2)\frac{\mu_S - z_{t_1}}{1 - t_1} = d_1 \frac{\mu_1 - z_{t_1}}{1 - t_1} + d_2 \frac{\mu_2 - z_{t_2}}{1 - t_2} \tag{33}$$

By expressing $z_{t_2} = z_{t_1} + d_1 \frac{\mu_1 - z_{t_1}}{1-t_1}$ and substituting it into the right-hand side, a remarkable mathematical simplification occurs: the current noisy state z_{t_1} cancels out entirely from both sides of the equation. Solving for the student's expected embedding μ_S, we obtain a strict convex combination of the teacher's expected embeddings:

$$\mu_S = \alpha\mu_1 + \beta\mu_2 \tag{34}$$

where the coefficients are defined exclusively by the time intervals:

$$\alpha = \frac{d_1}{d_1 + d_2}\frac{1 - t_3}{1 - t_2}, \qquad \beta = \frac{d_2}{d_1 + d_2}\frac{1 - t_1}{1 - t_2} \tag{35}$$

Because the expected embedding μ is a linear function of the predicted categorical probabilities q, Equation 34 implies that matching the continuous velocities in the embedding space is mathematically equivalent to matching a linear mixture of the categorical distributions:

$$q_{target} = \alpha q_1 + \beta q_2 \tag{36}$$

This insight is pivotal. It transforms a spatial regression problem into a classification problem. Instead of using a continuous MSE loss, which behaves poorly on categorical expectations, we optimize the student network using the KL divergence against the exact target distribution mixture:

$$\mathcal{L}_{\text{SVFM}} = D_{\text{KL}}(q_{target} \parallel q_{student}) \tag{37}$$

By decoupling the loss from the noise state z_{t_1} and penalizing only the distributional divergence, this formulation provides drastically improved training stability for discrete tokens.

E.2 Dual-EMA Bootstrapping

Distilling a multi-step trajectory into a few-step shortcut introduces a moving target problem: the student must align with a teacher for distillation, but must also align with its own future predictions to ensure trajectory self-consistency.

To stabilize this dynamic, we implement a Dual-EMA (Exponential Moving Average) bootstrapping strategy. We maintain two copies of the student model updated at different decay rates:

- **Fast EMA (Local Teacher):** Updated aggressively (decay $= 0.99$). It rapidly tracks the student's latest discoveries regarding the shortcut trajectory, serving as the immediate local target q_1 for self-consistency steps.
- **Slow EMA (Global Anchor):** Updated conservatively (decay $= 0.999$). It acts as the stable global target q_2 at t_2, preventing the student from collapsing into degenerate, high-error feedback loops.

During training, a fraction of the batch is strictly guided by the frozen pre-trained base model (pure distillation), while the remainder is guided by a mixture of the Fast and Slow EMA models (self-consistency).

F Experimental Insights: The Incompatibility of Continuous Distillation in Categorical Spaces

While the theoretical formulation of Shortcut Variational Flow Matching (SVFM) elegantly resolves the continuous velocity equation into a discrete probability mixture, empirical evaluation revealed profound structural incompatibilities. Over a series of experiments, we observed that applying continuous ODE-based distillation to categorical expected embeddings actively degraded generative performance. Notably, the base teacher model achieved a highly competitive 8-step FID of 24.77 natively, while the distilled student models consistently yielded worse FIDs (ranging from 40 to over 150). This section details the empirical progression, the attempted interventions, and the fundamental flaws uncovered in this paradigm.

F.1 The Entropy Collapse of Soft Mixtures

In our initial implementation, the student was trained to match the exact soft target distribution $q_{target} = \alpha q_1 + \beta q_2$ using the KL divergence. Empirically, we observed a deceptive training dynamic: the global loss reached a minimum around epoch 7 and subsequently began to rise, while the FID plateaued in the mid-40s. Analysis of the individual loss components revealed that while the distillation loss (anchored by the frozen teacher) remained stationary, the self-consistency loss exhibited a U-shape.

This behavior stems from the properties of convex combinations. The entropy of a strict convex combination of two disagreeing distributions is inherently greater than or equal to the average of their individual entropies. As the Slow and Fast EMA models iteratively fed blurred distributions back to the student, a feedback loop was established. The target distributions became increasingly uniform (high entropy), causing the student's categorical predictions to "smear" across the vocabulary. The rising loss was an artifact of the target's entropy increasing, preventing the model from resolving fine, high-frequency image details.

F.2 Hard Targets and the L2 Projection Trap

To combat the entropy collapse, we attempted to enforce zero-entropy targets. We calculated the continuous expected target vector $\mu_{target} = \alpha\mu_1 + \beta\mu_2$, projected it to the nearest valid VQ-VAE codebook token using L_2 distance, and utilized this hard pseudo-label for standard Cross-Entropy training.

This intervention led to a catastrophic degradation. While the total loss and prediction entropy plummeted (indicating the model had become highly confident) the images became nearly monochromatic, and the FID exploded to over 150. This failure highlighted a fundamental geometric property of the discrete latent space: embedding spaces are not convex. The mathematical average of two distinct visual token embeddings does not represent a semantic intermediate; rather, it yields a point floating in the "dead space" of the latent manifold. Projecting this floating point via L_2 distance arbitrarily snapped the targets to generic background tokens. By enforcing hard targets, we destroyed the uncertainty of the teacher, causing the student to overfit to trivial, monochromatic outputs.

F.3 The Fundamental Flaw: Off-Manifold Teacher Inputs

In a final attempt to resolve the trajectory math, we reverted to pure consistency distillation, forcing the student to match the target prediction exactly at the next timestep ($q_{target} = q_2$) to bypass the problematic α, β mixtures. However, this run also resulted in rising FIDs and oscillating losses, producing recognizable but severely detail-deficient images.

This revealed the ultimate structural flaw of applying SCFM to categorical flow matching: off-manifold stepping. In continuous diffusion, taking an intermediate Euler step $z_{t_2} = z_{t_1} + \Delta t \cdot v$ simply yields a slightly less noisy image. The teacher network evaluates this state perfectly. However, in our Variational framework, the velocity v is constructed using a dense mixture of codebook embeddings. Consequently, the intermediate state z_{t_2} becomes a continuous vector floating off the categorical data manifold.

When this off-manifold vector is fed into the frozen Transformer teacher (or the EMA models) to acquire the target q_2, the model's self-attention mechanism fails to properly resolve the unrecognizable input. The teacher becomes effectively "blind" and outputs a subdued, low-confidence prediction. We were mathematically forcing the student to learn from degraded targets, which explains why the distillation process actively eroded the teacher's native generation capabilities.

F.4 Conclusion and Future Directions

This experimental journey provides a robust negative result: continuous trajectory distillation algorithms, such as SCFM, are structurally incompatible with categorical latent spaces modeled via expected embeddings. The required continuous Euler integration forces the network to evaluate off-manifold states, triggering either entropy collapse or severe mode degradation.

Crucially, our baseline evaluations demonstrated that Variational Flow Matching natively learns remarkably straight trajectories. The pre-trained teacher achieved an FID of 21.2 in 250 steps and successfully maintained an FID of 24.7 in just 8 steps without any distillation.